7.4 - 17

Are the following sets of vectors linearly independent? Show the details of your work.

17. $[3 \quad 4 \quad 0 \quad 2]$, $[2 \quad -1 \quad 3 \quad 7]$, $[1 \quad 16 \quad -12 \quad -22]$

Let's create a matrix and reduce it

$$\begin{pmatrix} 3 & 4 & 0 & 2 \\ 2 & -1 & 3 & 7 \\ 1 & 16 & -12 & -22 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 1 & 16 & -12 & -22 \\ 2 & -1 & 3 & 7 \\ 3 & 4 & 0 & 2 \end{pmatrix} \xrightarrow[R_3 - 3R_1 \to R_3]{R_2 - 2R_1 \to R_2} \begin{pmatrix} 1 & 16 & -12 & -22 \\ 0 & -33 & 27 & 51 \\ 0 & -44 & 36 & 68 \end{pmatrix}$$

$$\xrightarrow{\frac{1}{-33}R_2 \to R_2} \begin{pmatrix} 1 & 16 & -12 & -22 \\ 0 & 1 & -\frac{9}{11} & -\frac{17}{11} \\ 0 & -44 & 36 & 68 \end{pmatrix} \xrightarrow[R_3 + 44R_2 \to R_3]{R_1 - 16R_2 \to R_1} \begin{pmatrix} 1 & 0 & \frac{12}{11} & \frac{30}{11} \\ 0 & 1 & -\frac{9}{11} & -\frac{17}{11} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The 3^{rd} row is zero, so I conclude:

The vectors are linearly dependents

7.4 - 18

Are the following sets of vectors linearly independent? Show the details of your work.

18. $[1 \ \ \frac{1}{2} \ \ \frac{1}{3} \ \ \frac{1}{4}]$, $[\frac{1}{2} \ \ \frac{1}{3} \ \ \frac{1}{4} \ \ \frac{1}{5}]$, $[\frac{1}{3} \ \ \frac{1}{4} \ \ \frac{1}{5} \ \ \frac{1}{6}]$, $[\frac{1}{4} \ \ \frac{1}{5} \ \ \frac{1}{6} \ \ \frac{1}{7}]$

Let's create a matrix to reduce it.

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \end{bmatrix} \xrightarrow[R_4 - \frac{1}{4}R_1 \to R_4]{R_2 - \frac{1}{2}R_1 \to R_2,\ R_3 - \frac{1}{3}R_1 \to R_3} \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ 0 & \frac{1}{12} & \frac{1}{12} & \frac{3}{40} \\ 0 & \frac{1}{12} & \frac{4}{45} & \frac{1}{12} \\ 0 & \frac{3}{40} & \frac{1}{12} & \frac{9}{112} \end{bmatrix} \xrightarrow{12R_2 \to R_2}$$

$$\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ 0 & 1 & 1 & \frac{9}{10} \\ 0 & \frac{1}{12} & \frac{4}{45} & \frac{1}{12} \\ 0 & \frac{3}{40} & \frac{1}{12} & \frac{9}{112} \end{bmatrix} \xrightarrow[R_4 - \frac{3}{40}R_2 \to R_4]{R_1 - \frac{1}{2}R_2 \to R_1,\ R_3 - \frac{1}{12}R_2 \to R_3} \begin{bmatrix} 1 & 0 & -\frac{1}{6} & -\frac{1}{5} \\ 0 & 1 & 1 & \frac{9}{10} \\ 0 & 0 & \frac{1}{180} & \frac{1}{120} \\ 0 & 0 & \frac{1}{120} & \frac{9}{700} \end{bmatrix} \xrightarrow{180R_3 \to R_3}$$

$$\begin{bmatrix} 1 & 0 & -\frac{1}{6} & -\frac{1}{5} \\ 0 & 1 & 1 & \frac{9}{10} \\ 0 & 0 & 1 & \frac{3}{2} \\ 0 & 0 & \frac{1}{120} & \frac{9}{700} \end{bmatrix} \xrightarrow[R_4 - \frac{1}{120}R_3 \to R_4]{R_1 + \frac{1}{6}R_3 \to R_1,\ R_2 - R_3 \to R_2} \begin{bmatrix} 1 & 0 & 0 & \frac{1}{20} \\ 0 & 1 & 0 & -\frac{3}{5} \\ 0 & 0 & 1 & \frac{3}{2} \\ 0 & 0 & 0 & \frac{1}{2800} \end{bmatrix}$$

All the rows have a pivot then I can conclude:

The Vectors are linearly independent

7.4 - 20

Are the following sets of vectors linearly independent? Show the details of your work.

20. [1 2 3 4], [2 3 4 5], [3 4 5 6], [4 5 6 7]

Let's create a matrix and reduce it!

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \end{bmatrix} \xrightarrow[R_4-4R_1\to R_4]{R_2-2R_1\to R_2,\ R_3-3R_1\to R_3} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & -2 & -3 \\ 0 & -2 & -4 & -6 \\ 0 & -3 & -6 & -9 \end{bmatrix} \xrightarrow{-R_2\to R_2} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & -2 & -4 & -6 \\ 0 & -3 & -6 & -9 \end{bmatrix}$$

$$\xrightarrow[R_4+3R_2\to R_4]{R_1-2R_2\to R_1,\ R_3+2R_2\to R_3} \begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are only 2 Pivots, then:

The vectors are linearly dependent!

7.4-32

Is the given set of vectors a vector space? Give reasons. If your answer is yes, determine the dimension and find a basis. ($v_1, v_2, \cdots$ denote components.)

32. All vectors in R^3 with $3v_1 - 2v_2 + v_3 = 0$, $4v_1 + 5v_2 = 0$

Let's create the corresponding matrix equation

$$\begin{bmatrix} 3 & -2 & 1 \\ 4 & 5 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

This is an homogeneous system

$$Ax = 0$$

Let's create the augmented Matrix and reduce:

$$\left[\begin{array}{ccc|c} 3 & -2 & 1 & 0 \\ 4 & 5 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{3}R_1 \to R_1} \left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{1}{3} & 0 \\ 4 & 5 & 0 & 0 \end{array}\right] \xrightarrow{R_2 - 4R_1 \to R_2}$$

$$\left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{1}{3} & 0 \\ 0 & \frac{23}{3} & -\frac{4}{3} & 0 \end{array}\right] \xrightarrow{\frac{3}{23}R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & -\frac{2}{3} & \frac{1}{3} & 0 \\ 0 & 1 & -\frac{4}{23} & 0 \end{array}\right] \xrightarrow{R_1 + \frac{2}{3}R_2 \to R_1}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & \frac{5}{23} & 0 \\ 0 & 1 & -\frac{4}{23} & 0 \end{array}\right]$$

This means

$$v_1 = -\frac{5}{23} v_3$$

$$v_2 = \frac{4}{23} v_3$$

Let $v_3 = 23t$

$$v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} -5t \\ 4t \\ 23t \end{bmatrix} = t \begin{bmatrix} -5 \\ 4 \\ 23 \end{bmatrix}$$

Then

$V = \text{span}\left\{ \begin{pmatrix} -5 \\ 4 \\ 23 \end{pmatrix} \right\}$ with $\begin{pmatrix} -5 \\ 4 \\ 23 \end{pmatrix}$ as the basis and $\dim(V) = 1$

7.4 -34

Is the given set of vectors a vector space? Give reasons. If your answer is yes, determine the dimension and find a basis. ($v_1, v_2, \cdots$ denote components.)

34. All vectors in R^n with $|v_j| = 1$ for $j = 1, \cdots, n$

At the first sight it is NOT a vector space.

- It does not include zero
- There are counter examples for the addition and scalar multiplication

Let

$$v = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{with } n = 2 \quad \text{and } c = 2$$

$$v + v = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \notin V$$

$$2v = 2\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \notin V$$

Then it is no a Vector Space

7.7 - 7

Showing the details, evaluate:

7. $\begin{vmatrix} \cos\alpha & \sin\alpha \\ \sin\beta & \cos\beta \end{vmatrix}$

$$\begin{vmatrix} \cos\alpha & \sin\alpha \\ \sin\beta & \cos\beta \end{vmatrix} = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

Now using the trigonometric identity

$$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

We can say

$$\begin{vmatrix} \cos\alpha & \sin\alpha \\ \sin\beta & \cos\beta \end{vmatrix} = \cos(\alpha+\beta)$$

7.7 - 12

Showing the details, evaluate:

12. $\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$

$$\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix} = a\begin{vmatrix} a & b \\ c & a \end{vmatrix} - b\begin{vmatrix} c & b \\ b & a \end{vmatrix} + c\begin{vmatrix} c & a \\ b & c \end{vmatrix}$$

$$= a(a^2 - bc) - b(ac - b^2) + c(c^2 - ab)$$

$$= a^3 - \underline{abc} - \underline{abc} + b^3 + c^3 - \underline{abc}$$

$$\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix} = a^3 + b^3 + c^3 - 3abc$$

7.7 -22

Solve by Cramer's rule. Check by Gauss elimination and back substitution. Show details.

22. $2x - 4y = -24$

$5x + 2y = 0$

Let's define the determinants

$D = \begin{vmatrix} 2 & -4 \\ 5 & 2 \end{vmatrix}$ $\quad D_x = \begin{vmatrix} -24 & -4 \\ 0 & 2 \end{vmatrix}$ $\quad D_y = \begin{vmatrix} 2 & -24 \\ 5 & 0 \end{vmatrix}$

$D = 2 \cdot 2 - 5 \cdot (-4)$ $\quad D_x = -24 \cdot 2$ $\quad D_y = -(5)(24)$

$D = 24$ $\quad D_x = -48$ $\quad D_y = 120$

then:

$x = \frac{D_x}{D}$ $\quad y = \frac{D_y}{D}$

$x = \frac{-48}{24}$ $\quad y = \frac{120}{24}$

$x = -2$ $\quad y = 5$

Lets check by Gauss reduction

$$\left[\begin{array}{cc|c} 2 & -4 & -24 \\ 5 & 2 & 0 \end{array}\right] \xrightarrow{\frac{1}{2}R_1 \to R_1} \left[\begin{array}{cc|c} 1 & -2 & -12 \\ 5 & 2 & 0 \end{array}\right] \xrightarrow{R_2 - 5R_1 \to R_2} \left[\begin{array}{cc|c} 1 & -2 & -12 \\ 0 & 12 & -60 \end{array}\right]$$

$$\xrightarrow{\frac{1}{12}R_2 \to R_2} \left[\begin{array}{cc|c} 1 & -2 & -12 \\ 0 & 1 & 5 \end{array}\right] \xrightarrow{R_1 + 2R_2 \to R_1} \left[\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & 5 \end{array}\right]$$

We got same results

7.8 - 2

Find the inverse by Gauss–Jordan (or by (4*) if $n = 2$). Check by using (1).

2. $\begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$

Lets do Gauss Jordan

$$\left[\begin{array}{cc|cc} \cos 2\theta & \sin 2\theta & 1 & 0 \\ -\sin 2\theta & \cos 2\theta & 0 & 1 \end{array}\right] \xrightarrow{\frac{1}{\cos 2\theta} R_1 \to R_1} \left[\begin{array}{cc|cc} 1 & \tan 2\theta & \sec 2\theta & 0 \\ -\sin 2\theta & \cos 2\theta & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 + \sin 2\theta R_1 \to R_2} \left[\begin{array}{cc|cc} 1 & \tan 2\theta & \sec 2\theta & 0 \\ 0 & \sec 2\theta & \tan 2\theta & 1 \end{array}\right] \xrightarrow{\frac{1}{\sec 2\theta} R_2 \to R_2} \left[\begin{array}{cc|cc} 1 & \tan 2\theta & \sec 2\theta & 0 \\ 0 & 1 & \text{Sen } 2\theta & \cos 2\theta \end{array}\right]$$

$$\xrightarrow{R_1 - \tan 2\theta R_2 \to R_1} \left[\begin{array}{cc|cc} 1 & 0 & \cos 2\theta & -\text{Sen } 2\theta \\ 0 & 1 & \text{Sen } 2\theta & \cos 2\theta \end{array}\right]$$

then the inverted matrix is:

$$\begin{bmatrix} \cos 2\theta & -\text{Sen } 2\theta \\ \text{Sen } 2\theta & \cos 2\theta \end{bmatrix}$$

7.8 -5

5. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 5 & 4 & 1 \end{bmatrix}$

Find the inverse by Gauss–Jordan (or by (4*) if $n = 2$). Check by using (1).

Lets do Gauss - Jordan

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 5 & 4 & 1 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3 - 5R_1 \to R_3]{R_2 - 2R_1 \to R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 4 & 1 & -5 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 - 4R_2 \to R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 3 & -4 & 1 \end{array}\right]$$

then the inverse is

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & -4 & 1 \end{bmatrix}$$

7.8 - 20

Formula (4) is occasionally needed in theory. To understand it, apply it and check the result by Gauss–Jordan:

20. In Prob. 6

6. $\begin{bmatrix} -4 & 0 & 0 \\ 0 & 8 & 13 \\ 0 & 3 & 5 \end{bmatrix}$

Let

$$A = \begin{bmatrix} -4 & 0 & 0 \\ 0 & 8 & 13 \\ 0 & 3 & 5 \end{bmatrix}$$

the formula 4 is

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A) \quad \text{where} \quad \operatorname{adj}(A) = [C_{jk}]^T$$

Lets compute the cofactor matrix

$$C_{11} = +\begin{vmatrix} 8 & 13 \\ 3 & 5 \end{vmatrix} = 40 - 39 = 1$$

$$C_{12} = -\begin{vmatrix} 0 & 13 \\ 0 & 5 \end{vmatrix} = 0$$

$$C_{13} = +\begin{vmatrix} 0 & 8 \\ 0 & 3 \end{vmatrix} = 0$$

$$C_{21} = -\begin{vmatrix} 0 & 0 \\ 3 & 5 \end{vmatrix} = 0$$

$$C_{22} = +\begin{vmatrix} -4 & 0 \\ 0 & 5 \end{vmatrix} = -20$$

$$C_{23} = -\begin{vmatrix} -4 & 0 \\ 0 & 3 \end{vmatrix} = 12$$

$$C_{31} = +\begin{vmatrix} 0 & 0 \\ 8 & 13 \end{vmatrix} = 0$$

$$C_{32} = -\begin{vmatrix} -4 & 0 \\ 0 & 13 \end{vmatrix} = 52$$

$$C_{33} = +\begin{vmatrix} -4 & 0 \\ 0 & 8 \end{vmatrix} = -32$$

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -20 & 12 \\ 0 & 52 & -32 \end{bmatrix}$$

$$\operatorname{Adj}(A) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -20 & 52 \\ 0 & 12 & -32 \end{bmatrix}$$

Now compute $\det |A|$

$$|A| = -4 \begin{vmatrix} 8 & 13 \\ 3 & 5 \end{vmatrix} = -4(40 - 39) = -4$$

Finally

$$A^{-1} = \frac{1}{4}\begin{bmatrix} 1 & 0 & 0 \\ 0 & -20 & 52 \\ 0 & 12 & -32 \end{bmatrix} = \begin{bmatrix} -\frac{1}{4} & 0 & 0 \\ 0 & 5 & -13 \\ 0 & -3 & 8 \end{bmatrix}$$

7.9-3

(More problems in Problem Set 9.4.) Is the given set, taken with the usual addition and scalar multiplication, a vector space? Give reason. If your answer is yes, find the dimension and a basis.

3. All vectors in R^3 satisfying $-v_1 + 2v_2 + 3v_3 = 0$, $-4v_1 + v_2 + v_3 = 0$.

Let's Create the corresponding matrix

$$\left[\begin{array}{ccc|c} -1 & 2 & 3 & 0 \\ -4 & 1 & 1 & 0 \end{array}\right]$$

this is homogenous, then is Vector Space

$$\left[\begin{array}{ccc|c} -1 & 2 & 3 & 0 \\ -4 & 1 & 1 & 0 \end{array}\right] \xrightarrow{-R_1 \to R_1} \left[\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ -4 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_2 + 4R_1 \to R_2}$$

$$\left[\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ 0 & -7 & -11 & 0 \end{array}\right] \xrightarrow{-\frac{1}{7}R_2 \to R_2} \left[\begin{array}{ccc|c} 1 & -2 & -3 & 0 \\ 0 & 1 & \frac{11}{7} & 0 \end{array}\right] \xrightarrow{R_1 + 2R_2 \to R_1} \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{7} & 0 \\ 0 & 1 & \frac{11}{7} & 0 \end{array}\right]$$

then $v_1 = -\frac{1}{7}v_3 \quad v_2 = -\frac{11}{7}v_3$

Let $v_3 = 7t$

$$v = \begin{bmatrix} -t \\ -11t \\ -7t \end{bmatrix} = t\begin{bmatrix} -1 \\ -11 \\ 7 \end{bmatrix}$$

Finally

$V = \text{span}\left(\begin{bmatrix} -1 \\ -11 \\ 7 \end{bmatrix}\right)$ where $\begin{bmatrix} -1 \\ -11 \\ 7 \end{bmatrix}$ is the basis

and $\dim(V) = 1$

7.9 - 4

(More problems in Problem Set 9.4.) Is the given set, taken with the usual addition and scalar multiplication, a vector space? Give reason. If your answer is yes, find the dimension and a basis.

4. All skew-symmetric 3×3 matrices.

A skew-symmetric matrix is

$$A^T = -A$$

then a generic formula of a 3×3 skew-symmetric matrix is

$$A = \begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix}$$

It can be witten like this:

$$A = a\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + b\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} + c\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

here we have a, b, c as parameters then

$$\dim(V) = 3$$

and

$$b_1 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad b_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix} \quad b_3 \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

are the elements in a basis of V

7.9 - 6

(More problems in Problem Set 9.4.) Is the given set, taken with the usual addition and scalar multiplication, a vector space? Give reason. If your answer is yes, find the dimension and a basis.

6. All functions $y(x) = a \cos 2x + b \sin 2x$ with arbitrary constants a and b.

We can think in

$\cos(2x)$ as an element in the basis

$\sin(2x)$ as another element in the basis

then to check if it is a Vector Space

$a \cos(2x) + b \sin(2x) = 0$

if $x = 0$

$a \cos(0) + b \sin(0) = 0$

$a + 0 = 0$

$a = 0$

If $x = \frac{\pi}{4}$

$a \cos\left(\frac{\pi}{2}\right) + b \sin\left(\frac{\pi}{2}\right) = 0$

$0 + b = 0$

$b = 0$